



National Health Claims Exchange

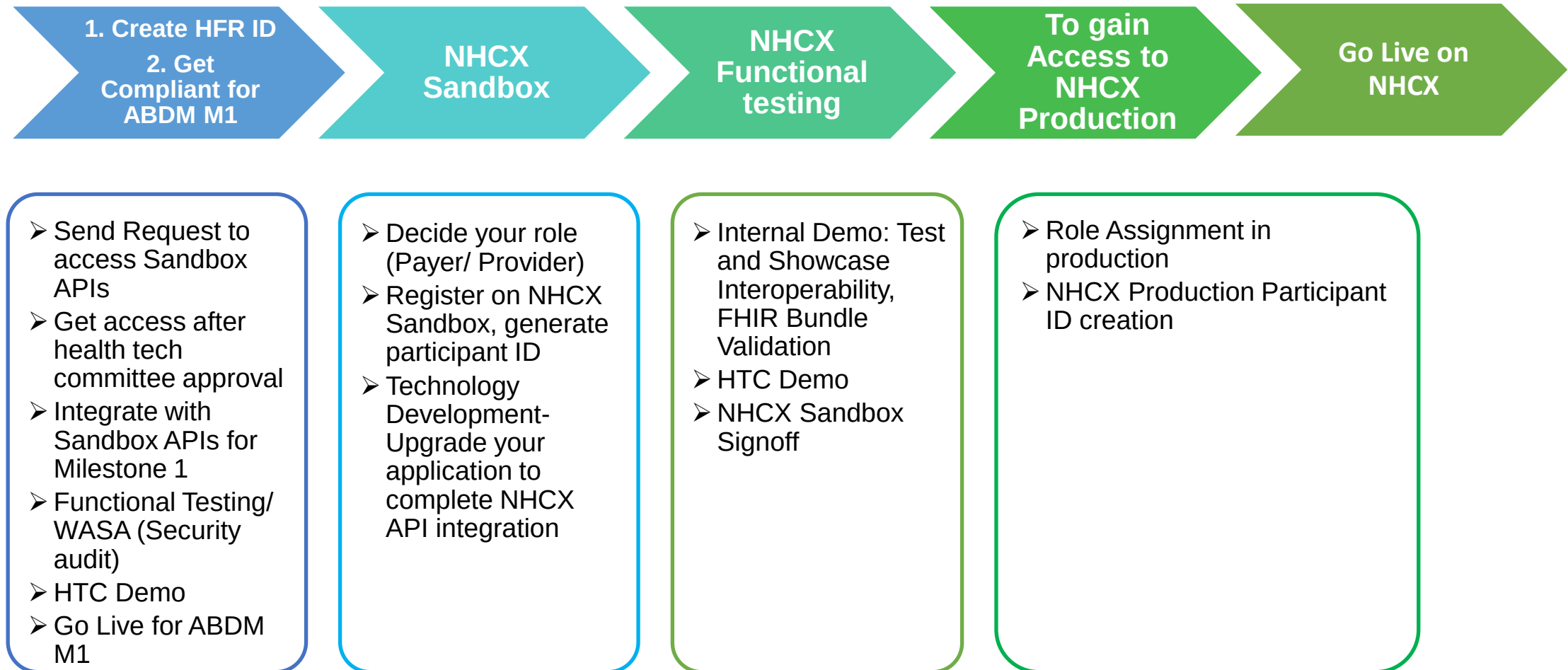
The next step in Digital Health Interoperability

Integrators Guide

Integrator's Journey



Guidelines for Participant Onboarding on NHCX



NHCX Pre-Requisites



Step 1: Registration with Health Facility Registry (HFR)

Please visit <https://facility.abdm.gov.in/> or <https://nhpr.abdm.gov.in/home> to complete the facility registration process. Please write to facility@nha.gov.in in case of any question on this step.

Step 2: Apply on ABDM sandbox by filling the registration form

The pre-requisite for NHCX is ABHA creation and verification. This implies that you must also **complete the milestone M1 integration process**.

Firstly, apply for sandbox registration : <https://sandbox.abdm.gov.in/sandbox/v3/>

In order to participate for NHCX integration select "**Providers and Payer**" and **M1** as an intent for request.

Once your application is reviewed and approved, you can begin your integration journey using the provided client ID and secret key .

Step 3: Successful M1 Integration under ABDM

Once your functional testing and WASA (security audit) are completed, an HTC demo will be conducted. Completing these steps will enable you to proceed to ABDM production.

Your software must have ABDM M1 functionalities enabled to proceed with NHCX integration.

HFR Registration



Our Toll Free number: 1800-11-4477/14477

+A A A- | English

Login/Registration Admin Login

Home About ABDM Resource Center Support Know Your Doctor/ Facility

Ayushman Bharat Digital Mission

Healthcare

Professionals Registry

Healthcare Professionals Registry (HPR) is a comprehensive repository of registered and verified different system of medicines (Modern medicine, Dentistry, Ayurveda, Unani, Siddha, Sowa-Rigpa, Homeopathy) and Nurses practitioners delivering healthcare services across India.

[Read more](#)

Health

Facility Registry

Health Facility Registry is a comprehensive repository of health facilities of the country across modern and traditional systems of medicine. It includes both public and private health facilities including hospitals, clinics, diagnostic

[Read more](#)

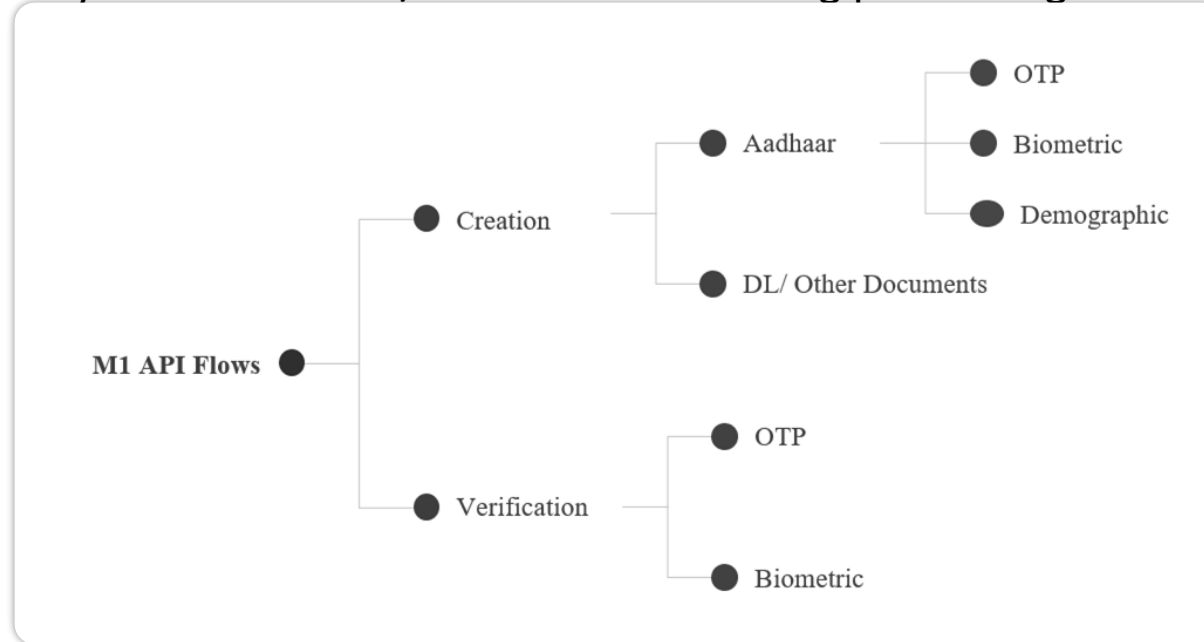
1. To login or register in HFR, click on “Login/Registration.”
2. **For detailed guide please refer to the below link: [HFR creation](#)**

ABDM M1 Integration



Functionalities to be implemented as part of M 1

1. Create ABHA number using Aadhaar / Driving License
2. Verify ABHA Number / ABHA address during patient registration



Documentation on ABDM M1

- <https://sandbox.abdm.gov.in/sandbox/v3/new-documentation?doc=getting-started>
- https://sandbox.abdm.gov.in/sandbox/v3/new-documentation?doc=Milestone_one
- https://sandbox.abdm.gov.in/sandbox/v3/new-documentation?doc=postman_collections
- https://sandbox.abdm.gov.in/sandbox/v3/new-documentation?doc=ABDM_M3_V1

Please follow our test cases for M1

<https://sandbox.abdm.gov.in/sandbox/v3/new-documentation?doc=TestCases>

Note: Integration support team will guide you if you encounter any roadblocks in the process. Please reach out to integration.support@nha.gov.in

Integrator's Journey (Sandbox Participant Creation)



Guidelines for Participant Onboarding on NHCX



1. Register on the Sandbox

- Click on the following link to register: <https://sandbox.abdm.gov.in/sandbox/v3/sandbox-registration>
- Use your ABDM sandbox client ID and secret key during the registration process.

Valid Role Enums :
PROVIDER("10001")
PAYER("10002")
AGENCY_TPA("10003")
AGENCY_REGULATOR("10004")
RESEARCH("10005")

2. Obtain Credentials

After submitting the registration form, you will be assigned roles to access the NHCX Sandbox environment.

3. Generate Your Participant ID

- Use the client ID and client secret obtained from the ABDM sandbox portal during Milestone 1 integration to generate an access token.
API to generate access token: <https://dev.abdm.gov.in/gateway/v0.5/sessions>
- Pass the access token in the headers of the following API: - **API to create participant:** <https://apisbx.abdm.gov.in/pmjay/sbxhcx/participantnchxservice/participant/create> API
- Use the same credentials to create participant code for all the facilities using their respective HFR ID."

Valid Registry Enums:
HFR ("10001")
NIN ("10002")
ROHINI(" 10003")
PAYER ("10004")

Integrator's Journey (Technology Upgradation)



Guidelines for Participant Onboarding on NHCX



1. Base URL for Sandbox:

Use Case APIs(useCaseBaseUrl)- <https://apisbx.abdm.gov.in/hcx>

2. Please visit <https://hcxsbx.abdm.gov.in/#/documents>. The use case documents for providers and payers are available on the website and can be downloaded.

Integrator's Journey



Guidelines for Participant Onboarding on NHCX



1. FHIR Bundle Validation:

Please send the FHIR bundles to hcx.integration@nha.gov.in for validation by the NRCeS Team.

2. Functional Testing:

- a. **Internal Demo:** NHA team will be conducting this demo to check the use cases
- b. **HTC demo:** This will be conducted by NRCeS, IRDAI, TCS and NHA teams

Please Note: to request for the demos, the integrators will have to write an email on the above-mentioned email

3. After **successfully completing the demos and FHIR bundle validation**, you will receive a communication from NHA confirming your successful integration on the NHCX sandbox.

Integrator's Journey



Guidelines for Participant Onboarding on NHCX



- After successful integration, **NHA team will assign provider role to your M1 production clientID and same will be used for NHCX production access.**
- **Base URL for production: Will be shared post successful integration on NHCX Sandbox**
- Use this link for onboarding APIs on NHCX production and usecases: <https://nhcx.abdm.gov.in/>

Please refer this link for detailed documentation on Production: <https://nhcx.abdm.gov.in/>

Integrator's Journey



Guidelines for Participant Onboarding on NHCX



1. Participants must prepare their application for go-live in their respective environments after obtaining the production credentials.
2. NHCX recommends conducting necessary training for staff and planning for change management before going live in production, preferably after a pilot with a small set of clients.

Error Handling



Acceptance scenario

Whenever recipient receives a request from the sender (through NHCX) the return type should be as below with http status as 202 Accepted.

```
{
  "timestamp":"DD/MM/YYYY hh:mm:ss:sss",
  "api_call_id":"UUID",
  "correlation_id":"UUID",
  "result":{
    "sender_code":"PYRXX@hcx",
    "recipient_code":"INXXXXXX@hcx",
    "entity_type":"coverageeligibility/preauth/claim/task/payment/insuranceplan",
    "protocol_status":"request.queued/request.dispatched/request.error"
  },
  "error":{
    "code":"",
    "message":""
  }
}
```

Error Handling



Error Scenario

If the recipient rejects the payload sent by the sender for any reason, or if the required format is not followed, the system will treat it as an error. In such cases, the system will attempt to resend the request up to five times before ultimately terminating the entire request.

After five failed attempts, the request associated with the specific correlation ID will be deleted from the NHCX system.

To avoid issues, the receiver should correctly follow the acceptance process or send an appropriate response if rejecting the payload. If the receiver fails to handle the error properly or if there is a server-side issue, the request will be returned to the sender.

So, every integrator should implement the [v1/error API](#) at their end where they will get the reject details and respond back.

Protocol Response in case of error scenarios to be sent back by the receiver in case of not acceptance after evaluation:

```
{
  "type": "ProtocolResponse",
  "x-hcx-sender_code": "",
  "x-hcx-recipient_code": "",
  "x-hcx-api_call_id": "UUID",
  "x-hcx-correlation_id": "UUID",
  "x-hcx-workflow_id": "UUID",
  "x-hcx-timestamp": "",
  "x-hcx-debug_flag": "Error",
  "x-hcx-status": "response.error",
  "x-hcx-redirect_to": "",
  "x-hcx-error_details": {
    "code": "String",
    "message": "String",
    "trace": "String"
  },
  "x-hcx-debug_details": {
    "code": "String",
    "message": "String",
    "trace": "String"
  },
  "x-hcx-domain-header": {
    "use_case_name": "String",
    "amt_processed": "String"
  },
  "x-hcx-entity-type": "coverageeligibility | payment | insuranceplan | task | claim | preauth",
  "x-hcx-ben-abha-id": "abha number"
}
```

Important Links and URLs



1. Website Links:

Sandbox: <https://hcxsbx.abdm.gov.in/>

Production: <https://nhcx.abdm.gov.in/>

2. Base URL for Sandbox:

Participant APIs(participantBaseUrl)-

<https://apisbx.abdm.gov.in/pmjay/sbxhcx/participanthcxservice>

Use Case APIs(useCaseBaseUrl)- <https://apisbx.abdm.gov.in/hcx>

3. Base URL for production:

Participant APIs(participantBaseUrl)-

<https://apisprod.nha.gov.in/pmjay/hcx/participanthcxservice>

UseCase APIs(useCaseBaseUrl)- <https://apisprod.nha.gov.in/hcx>

4. Please refer to the documents here:

<https://nhcx.abdm.gov.in/#/documents>

5. Onboarding Documents:

[Onboarding providers and payers in Sandbox](#)

[Onboarding providers and payers in Production](#)

6. Use Case Documents:

[NHCX Payer Side Use Cases- Sandbox Exit Process](#)

[NHCX Provider Side Use Cases- Sandbox Exit Process](#)

- The passcode will be sent the registered number after successful validation and the transaction id will be shared in the response; both can be further used for confirmation.

Refer the NRCES technical specifications for the respective use case . For any query regarding this , please connect with NRCES team at mail id- nrc-help@cdac.in.



Thank You